SAMUEL KAZ

Glen Rock, NJ — samuel.kaz1@gmail.com — 201-364-4242

SKILLS:

Technical Skills: ISO 13485:2016 Documentation, Project Management, CAD/CAE, Robotics
Software: SolidWorks, Draftsight, MATLAB, CREO, Arena, Arduino, Minitab, Visual Studio, MS Office Suite
Programming: Arduino, MATLAB, C++

PROFESSIONAL EXPERIENCE:

Design Engineer | Mainstream Fluid & Air, Berkeley Heights, NJ — Jun 2025-Jun 2026

- Engineered hundreds of custom fan arrays to retrofit AHUs, optimizing reliability and design output quality.
- Performed airflow, pressure, and spatial analysis to ensure effectiveness, increasing efficiency by 30%.
- Generated assembly drawings and unit specifications using Draftsight to streamline RFQ approval.

Systems Integration Lab Graduate Researcher and TA | Stevens, Hoboken, NJ — May 24-May 2025

- Developed and refined an ARUCO robotic boat used to autonomously locate and retrieve objects.
- Implemented an MPU accelerometer integrated with a PID controller using Arduino for automatic maneuvering.
- Mentored 60+ students through project development, providing baseline code and robot prototypes for support.

Fiber Engineering Intern | PPL Corporation, Allentown, PA — May 2023-Aug 2023

- Developed a resource website on fiber optic and substation integration, achieving an 80% adoption rate.
- Quickly learned and utilized company-specific software to complete Splice plans for fiber infrastructure.

CAD Engineering Intern | H2 Energy Now — Jun 2022-Aug 2022

- Utilized CAD software to blueprint a 12 component assembly of a patented Hydrogen Separation Chamber.
- Presented research and proposed design options regarding the best suited materials and fabrication methods.

PROJECTS:

Medical Device Design and Manufacturing Projects — Oct 2024-Apr 2025

- Conducted detailed DFM and DFA assessments via disassembly of medical devices for technical analysis, centering design recommendations on risk management, design controls, sterility, and biocompatibility.
- Analyzed medical device design and manufacturing processes in accordance with ISO 13485:2016 and FDA 21 CFR Part 820, authoring product development plans and a regulatory compliance report for product packaging.
- Evaluated theoretical Process Validation frameworks and applied FMEA to mitigate device failure modes.

Earthworm Soft Body Robot (CAPSTONE): — Sep 2023-May 2024

- Led a team to engineer a medical prototype capable of traveling within the brain to perform thrombectomy, reducing required procedural staff by 50% and lowering treatment costs for hospitals.
- Conducted rigorous research and technical analysis in Nitinol actuation, manufacturing, and biocompatibility.
- Followed industry product development phases, producing pitch-ready deliverables for investors.

Automatic Pet Feeder: — Mar 2023-May 2023

- Project coordinator and CAD modeler for a cost-effective automatic food, water and toy dispensing robot.
- Programmed microcontrollers to regulate various actuating systems based on environmental changes.

EDUCATION:

Stevens Institute of Technology | Hoboken, NJ

Master of Engineering in Mechanical Engineering, Concentration in Medical Devices — Graduated May 2025

GPA: 3.9/4.00 | **Coursework:** Medical Device Design/Manufacturing, Electron Microscopy, Mechatronics

Bachelor of Engineering in Mechanical Engineering, Minor in Electrical Engineering — Graduated May 2024

Coursework: Fluid Dynamics, Microprocessor Systems, Control Systems, Thermal Engineering, Electronic Circuits
Honors: Degree with Honor, Dean's List, 2nd place in Gallois Autonomous Robot Competition

INTERESTS:

Robotics, Chess, Varsity Tennis, Skiing, Ping Pong, 3D Modeling, Ultimate Frisbee